# Smart Energy Microgrid & Demand Forecasting Engine

**A Zoho Catalyst and QuickML project for forecast-led microgrid operations.**

# Microgrids Balance Three Competing Priorities

## Demand uncertainty makes energy purchasing and dispatch decisions harder.

Grid disruptions can force rapid transitions between grid, storage, and local generation. Static schedules react too late when conditions change.

### Affordability

Buy and dispatch energy at the lowest workable cost.

### Resilience

Shift quickly between grid, storage, and local generation.

### Critical-load continuity

Keep essential loads powered through every disruption.

# One Flow Connects Forecasts To Dispatch

**Inputs**

Interval demand, weather signals, tariffs, battery state, and grid-status signals feed the flow.

**Catalyst orchestration**

Zoho Catalyst ingests events, runs orchestration, and exposes the application workflow.

**QuickML forecast**

QuickML produces demand forecasts and confidence-aware decision inputs for dispatch.

**Dispatch and audit**

Recommends operating modes and records every decision for review.

# QuickML Makes Demand Operational

## Training and validation

QuickML trains on historical demand patterns and relevant external features. Validation separates training observations from held-out periods.

## Controlled handoff

Forecast error is reviewed before recommendations reach the transition engine. The model produces a near-term demand signal, not an autonomous production dispatch command.

# Safety Rules Govern Every Transition

**Operating inputs**

Forecast demand, battery state, tariff conditions, and grid status feed the decision.

**Recommendation**

Inputs determine the recommended mode.

**Critical loads first**

Priority holds during constrained or disrupted conditions.

**Unsafe transitions blocked**

Switching requires a valid operating state.

**Validation caveat**

Zero blackout risk awaits controlled testing.

# An Editorial Terminal Explains Decisions

## Visible Operating Context

The terminal presents current load, forecast direction, operating mode, and recommended action. Operators can inspect why the system recommends charging, discharging, importing, or conserving.

## Reviewable Decisions

A concise decision log supports review after each transition. The interface favors legible operational context over black-box automation.

# Tests Map Directly To The Rubric

## Forecasting

- Demand predictions are validated against held-out observations.
- Held-out testing shows how the forecast performs on data it never saw.

## Transitions

- Tests verify that allowed mode changes run as specified.
- Unsafe states are confirmed as blocked across every tested transition.

## Experience

- Checks confirm terminal clarity and decision traceability.
- Operator comprehension is tested so system behavior stays understandable.

## Submission alignment

- The project claims 100% alignment with the rubric.
- Each test area maps directly to a rubric criterion for review.

# Forecast-Led Control Creates Practical Value

## 22% reduction

Stated project claim for energy costs through forecast-led dispatch

## Resilience

Critical-load-aware transitions support continuity planning during disruptions

## Operations

Explainable recommendations reduce the burden of manual interpretation

## Grid value

Flexible demand and storage coordination can improve local balancing

# Prototype To Controlled Pilot

**Hackathon prototype**

Demonstrate ingestion, forecasting, terminal UX, and simulated dispatch.

**Validation phase**

Expand scenarios, tune guardrails, and document exceptions.

**Controlled pilot**

Connect approved data feeds and human-reviewed operating procedures.

**Scale phase**

Add site-specific models, reporting, and governance controls.

# HOUSTATLANTAVEGAS turns demand forecasts into explainable microgrid operating recommendations.

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